F Microwave Absorbers

In some of the experimental modules microwaves are used to drive transitions. This radiation needs to be contained so it does not interfere with other parts of the experiment. In order to do so, we considered covering the interior metal surfaces of some of the vacuum chambers with absorbing materials. In the following, I present several types of radiation-absorbing sheets, then evaluate how effective this approach can be at localizing the radiation.

F.1 Simple absorbers

Thin-film absorber In the simplest case, radiation is normally incident on an infinite resistive sheet of resistivity ρ. In terms of the parameter $x = Z_0 t/(2\rho)$, where $Z_0 \approx 376.73\ \Omega$ is the impedance of free space, the fraction of power transmitted, reflected, and absorbed is [138, eqn 8.79]

$$P_\mathrm{t} = \frac{1}{(1+x)^2}, \quad P_\mathrm{r} = \frac{x^2}{(1+x)^2}, \quad P_\mathrm{a} = \frac{2x}{(1+x)^2}.$$

These expressions are plotted in Fig. 101, where it is evident that the resistive sheet can absorb at most half of the power when $x = 1$, i.e. when the sheet thickness is $t = 2\rho/Z_0$. For gold ($\rho \approx 2.4 \cdot 10^{-8}\ \Omega \cdot \mathrm{m}$), this corresponds to a thickness of $t \approx 1$Å, which would be thinner than a sheet a single atom thick. In other words, this approach requires a material with a higher resistivity, and even then offers only 3 dB of absorptive attenuation.

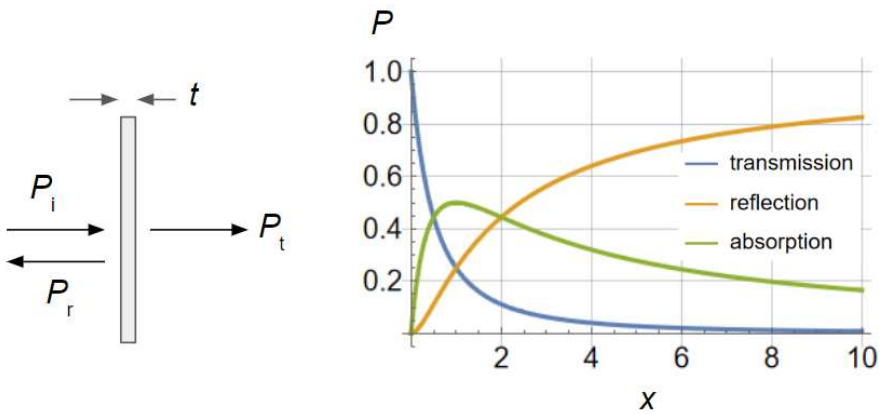

Figure 101: Thin-film absorber. The plot shows the fraction of transmitted (P_t), reflected (P_r), and absorbed (P_a) power as a function of the parameter $x = Z_0 t/(2\rho)$.

Two thin films between a substrate A much more flexible approach is to use two conductive (or resistive) sheets separated by a dielectric substrate of refractive index n and thickness h. The resistivity of each sheet is given as "ohms per square", reflecting the fact that the resistance between two opposite sides of a square piece of material of a given thickness and bulk resistivity is independent of the size of the square. The absorption of incident radiation (vacuum wavelength λ_0) can be calculated with the usual Fabry–Pérot methods and is [139]

$$A = \frac{4pq^2(f+f') + 4pf(p+f'-q)(p+f'+q)\sin^2\beta}{q^2(2p+f+f')^2 + (p+f-q)(p+f+q)(p+f+q)(f+f'-q)(p+f'+q)\sin^2\beta},$$

where

$$\begin{aligned}
f &= Z_0/R_\square = \text{reciprocal normalized surface resistance of left surface} \\
f' &= Z_0/R'_\square = \text{reciprocal normalized surface resistance of right surface} \\
p &= \cos\theta \\
q &= n\cos\theta_2 \\
\theta &= \text{angle of incidence} \\
\beta &= (2\pi nh\cos\theta_2)/\lambda_0 \\
\cos\theta_2 &= 1 - n^{-2}(1-\cos^2\theta).
\end{aligned}$$

Perfect absorption ($A = 1$) obtains in the case of the so-called "Salisbury screen', when one of the sheets has $F = Z_0/R_\square = 1$ (nicknamed "space cloth" since the resistivity equals the impedance of free space), while the other is perfectly conductive ($f' = \infty$). The separation between the two sheets needs to be

$$\beta = \frac{2\pi nh\cos\theta_2}{\lambda_0} = \frac{\pi}{2} \quad \rightarrow \quad h = \frac{\lambda_0}{4n}.$$

At 26 GHz, the quarter wavelength is $\lambda_0/4 \approx 2.89$ mm, showing that this approach could be implemented in practice. Fig. 102 shows the result for a HFSS simulation of a finite size absorber of this type; note how the absorption as a function of sheet spacing shows

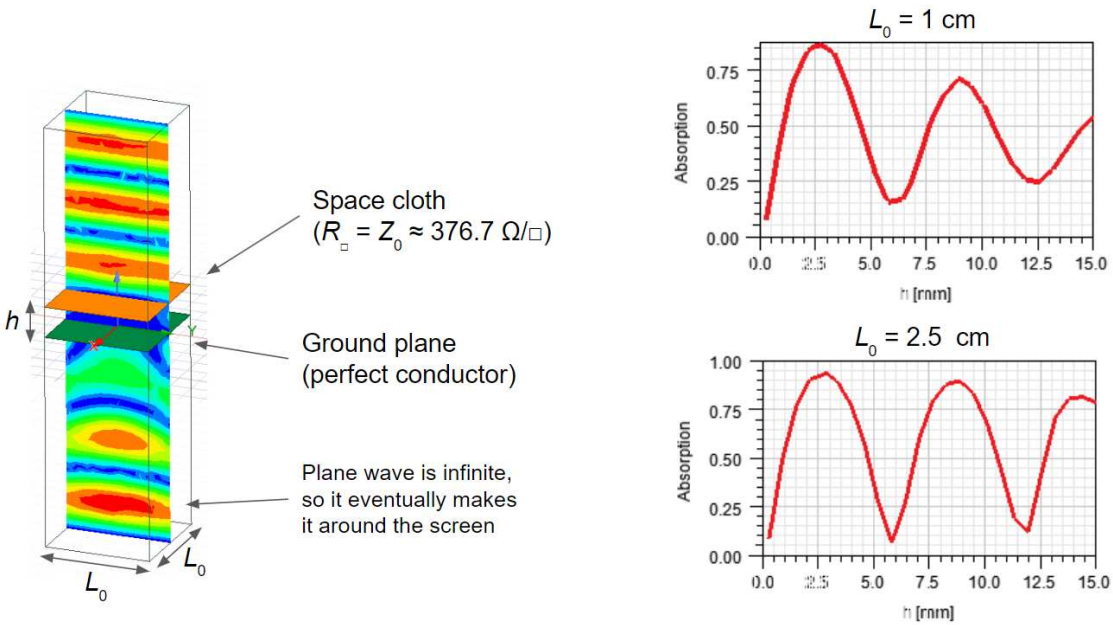

Figure 102: Microwaves incident onto a finite-sized "Salisbury screen", showing the dependence of absorption as a function of separation between the perfect conductor and the "space cloth" of precisely tuned sheet resistivity. Note how the larger sample shows less of a decay of the sinusoid in the absorption plot; and infinite sheet would be perfectly sinusoidal.

a decaying sinusoidal pattern. (An infinite-sized absorber, as simulated using periodic boundary conditions, would not exhibit the decay.)

Unlike the single-sheet absorber considered previously, the Salisbury screen is frequency selective as would be expected for a Fabry–Pérot structure. In general, we expect the transmission or absorption to be periodic in wavelength or cavity length. The periodicity has a length scale known as the free spectral range. Between two spectral peaks, or between zero (say, in cavity length) and the and the first peak, there are no additional peaks. In other words, no resonance occurs if the cavity (or spacing between the two sheets) is much shorter than the wavelength of the incident radiation.

For Fabry–Pérot absorbers (or cavities in general), the spectral features have a width determined by the reflectivity of the two mirrors, as characterized by the finesse $F = \pi\sqrt{R}/(1-R)$, where $R = \sqrt{R_1 R_2}$ is the geometric mean of the two reflectivities. In the

case of the Salisbury screen, we may assume

$$R_1 = \text{approx. 1 to 10, depending on } n$$

$$R_2 = 1, \text{ i.e. a perfect metallic mirror.}$$

Since one of the "mirrors" is determined by the air–substrate interface, it has a poor reflectivity, which limits the finesse of the cavity to some small number. Low-finesse cavities have very broad peaks, so there is some frequency dependence even for ultra-short cavities.

The two-sheet absorber follows the Fabry–Pérot theory. In the following paragraph, we consider a very different approach to constructing radiation-absorbing sheets which relies on resonances within planar structures rather than cavity-type resonances.

F.2 Metamaterial absorbers

In a typical microwave metamaterial absorber (see Fig. 103 for examples), a simple pattern (typically, circles, crosses, lines, rectangles) is repeated many times on a printed circuit board. On the absorption spectrum, the peaks are typically very sharp. To achieve effective absorption at several frequencies, the designers can choose patterns with several length scales (such as several nested rings [140], where the dimensions of each ring can be tuned to match one of the frequencies of interest). For a broadband absorber, patterns need to have a range of different length scales such that each feature's absorption spectrum partially overlaps with another, slightly bigger or smaller, feature (see [141] for a good review of the topic). In the following, I will discuss three ways of looking at metamaterial absorbers: as Fabry–Pérot resonators, as microstrip resonators, and as a phased antenna array.

Fabry–Pérot theory A metamaterial absorber consists of a ground plane (assumed perfectly conductive), a substrate, and a patterned surface. It is thus very similar to the two-sheet absorbers considered in the previous section, such as the Salisbury screen, except that the patterned surface itself has a complicated frequency dependence. Applying the standard Fabry–Pérot approach, which they call "multi-reflection interference theory",

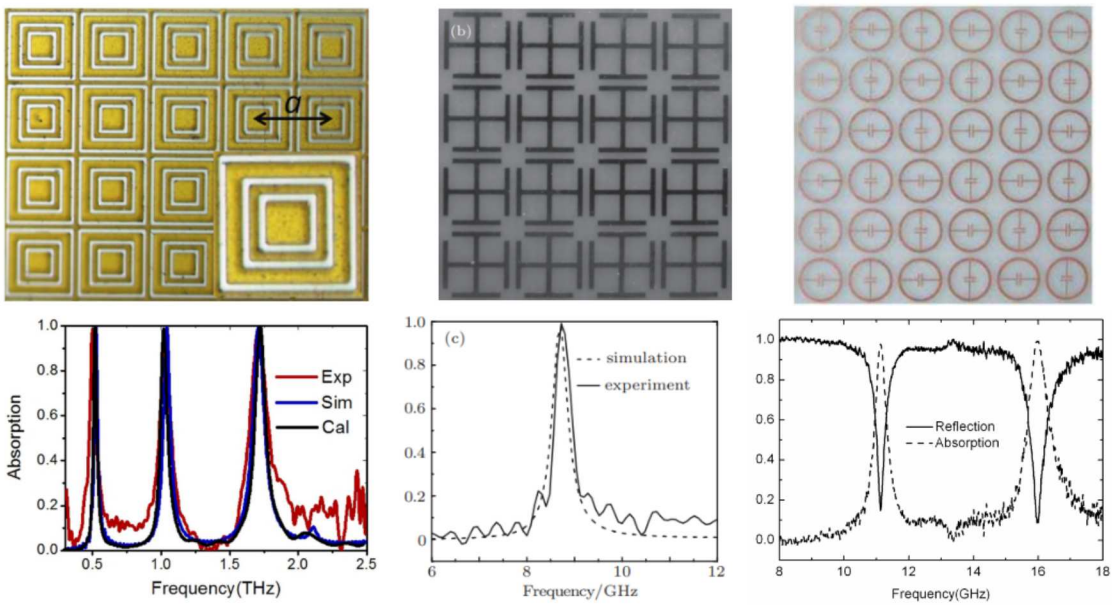

Figure 103: Examples of metamaterial absorber patterns (not to scale) with their corresponding absorption spectra. Figures taken from, respectively: [140], [142], [143]

Chen et al. [144] obtain the reflection coefficient for a metamaterial absorber,

$$\tilde{r} = \tilde{r}_{12} - \frac{\tilde{t}_{12}\tilde{t}_{21}e^{i2\tilde{\beta}}}{1 + \tilde{r}_{21}e^{i2\tilde{\beta}}},$$

where $\tilde{r}_{12}, \tilde{r}_{21}, \tilde{t}_{12}, \tilde{t}_{21}, \tilde{\beta}$ are numerically calculated and highly frequency dependent. While the authors advertise their "profound understanding of the underlying physics" in coming up with a theory that "explains all features observed in narrowband metamaterial absorbers", the 'profound physics' actually takes place in the numerically calculated coefficients which encode the resonances that take place within the patterned surface rather than the substrate-filled cavity formed by the patterned surface and the ground plane.

From their simulations, the authors show a plot of absorption spectrum as a function of the cavity length, where the resonant frequency changes by approx. 20% despite scanning the cavity length by a factor of 4, providing further clues as to the inner workings of a metamaterial absorber. In the following, I will show what I consider more enlightening ways to understand the underlying physics.

Microstrip resonators A microstrip transmission line consists of a conductor of width W placed over a ground plane separated by a substrate of permittivity ϵ_r and thickness d. The characteristic impedance of such a transmission line depends on the dimensions of the conductor (specifically, the ratio of the width of the line to the substrate height):

$$Z_0 = \begin{cases} \dfrac{60}{\sqrt{\epsilon_\mathrm{e}}} \ln\left(\dfrac{8d}{W} + \dfrac{W}{4d}\right) & \text{for } W/d \leq 1 \\ \dfrac{120\pi}{\sqrt{\epsilon_\mathrm{e}}[W/d + 1.393 + 0.667\ln(W/d + 1.444)]} & \text{for } W/d \geq 1, \end{cases}$$

where ϵ_e is the effective permittivity, which also depends on the dimensions of the transmission line [3, eqns 3.195–6].

$$\epsilon_\mathrm{e} = \frac{\epsilon_\mathrm{r} + 1}{2} + \frac{\epsilon_\mathrm{r} - 1}{2} \frac{1}{\sqrt{1 + 12d/W}}.$$

The effective permittivity is intermediate between the permittivity of air and the substrate itself, reflecting the fact that half of the line is surrounded by the substrate, and half by air.

It is possible to calculate the attenuation of a signal traveling down a microstrip transmission line due to dielectric as well as conductor-resistive losses [3, eqns 3.195–6] and on most substrates, the conductor loss is more significant than the dielectric loss except for some semiconductor substrates. [3]

As with any kind of transmission line, a length of a microstrip transmission line forms a resonator with resonant frequencies approximately determined by the line length, and the Q factor determined by the various transmission line losses. For example, in a microstrip ring resonator (see Fig. 104, as well as [3, problem 6.16]), the resonant frequencies are

$$f = \frac{nc}{2\pi a\sqrt{\epsilon_\mathrm{e}}}.$$

The connection to metamaterial absorbers is obvious; the absorption peaks correspond to the resonant frequencies of the microstrip resonator.

The part of the energy that is not dissipated in the conductor and substrate losses is radiated back into space. In fact, connecting a microstrip resonator to an active device is a space- and cost- effective way of creating an antenna. “Microstrip antennas are planar

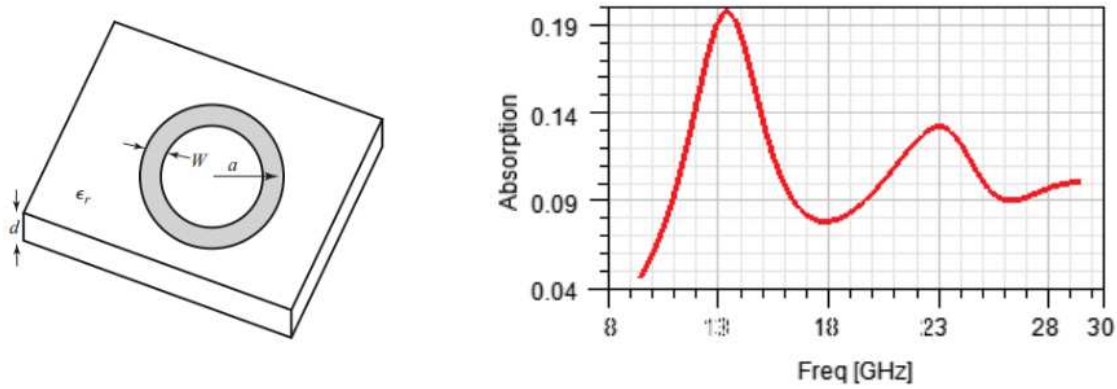

Figure 104: Microstrip ring resonator, $a = 2$ mm, $W = 0.3$ mm, $\epsilon_r = 4.4$, $d = 2.88$ mm, $f = 11.4$ GHz. NB: no periodic boundary conditions! Single cell only. ***Left.*** Geometry of the problem, from [3]. ***Right.*** HFSS simulation results.

resonant cavities that leak from their edges and radiate." [145] The ease of fabrication facilitates the assembly of an array of such antennas, which we discuss in the following paragraph.

Phased antenna arrays The microstrip patch antenna has a relatively wide radiation pattern on account of its small size. Assembling an array of such antenna elements and driving it with a fixed phase relationship between the antenna elements creates a phased antenna array which has a response similar to a diffraction grating (Fig. 105). The greater the number of antenna elements, the sharper the peaks in the spectrum. The peak spacing in terms of wavelength is proportional to the reciprocal antenna element spacing.

By adjusting the phase of the signal received by each antenna element, the antenna

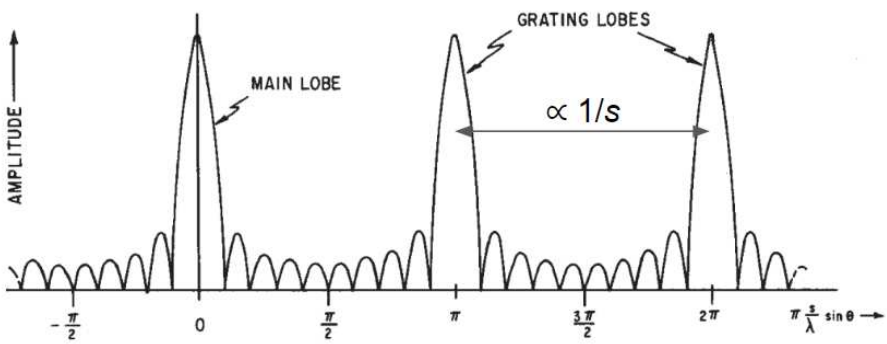

Figure 105: Response of an antenna array as a function of element spacing s, wavelength λ, and incident angle θ. Adapted from [146].

beam can be steered electronically. Considering a metamaterial absorber as a phased antenna array, the phase of each element is arbitrary, therefore a metamaterial absorber can effectively absorb energy from a wide range of angles.

Design process The key points to remember about metamaterial absorbers are that the frequency dependence is mostly driven by the dimensions of the unit cell, as well as the unit cell spacing. With that in mind, we can propose the following design algorithm:

1. Choose element shape and substrate material (given by ϵ_{r} and $\tan\delta$) and thickness.
2. Tune the basic element for maximum absorption at a given frequency.
3. Tune element spacing so grating peaks correspond to absorption peaks of the basic element.
4. Tune the unit cell again, and iterate as needed.

Building on this basic procedure, there are a number of design aspects to explore:

- Multi-frequency design (e.g., nested rings)
- Is it possible to invert patterns? E.g. a sheet with holes rather than an array of circles.
- What is the relative significance of ohmic vs dielectric losses? Is there a way to tune conductor or substrate properties to maximize absorption?
- What is the angular dependence of absorption, and to what extent is it relevant in a given application?
- What is the sensitivity to manufacturing tolerances? Is it necessary to experimentally determine the optimal dimensions to obtain the desired frequency response?
- What is the best way to measure the absorption of a given material? (E.g., use the same antenna to receive and transmit signals; have two separate antennas to receive and transmit, at a small angle to each other; use two distinct antennas but use a beam splitter to separate the forward and backward signals; place the metamaterial in a waveguide and measure absorption, etc.)

- Rather than using microstrip resonators, would it be better in any way to use other technologies, e.g. an array of dielectric resonators used as a metamaterial?

Rather than pursue these lines of research further, developing a good metamaterial absorber, in the following section I consider how much of an improvement the availability of such an absorber would be for the CENTREX experiment.

F.3 Expected effectiveness

Literature about microwave absorbers states the effectiveness of the absorbers in terms of an absorption coefficient for a plane wave normally incident upon an infinite absorbing sheet. This can be further refined to study the angular dependence, the effect of beam curvature for a Gaussian beam, the effects of the finite size of the absorbing material, however the problem we are trying to solve in CENTREX is somewhat different. We use a beam-shaping horn in an attempt to transmit a Gaussian beam of microwaves through a vacuum chamber. Specifically, the beam enters through a window on one side of the chamber, and would ideally exit the chamber through the opposing window without reflections or scattering that would give rise to appreciable microwave power in areas outside the main microwave beam.

In practice, the beam will get scattered from the windows, the walls of the chamber, and the electrodes and other instrumentation present within each vacuum chamber. The question is whether covering the interior walls of the vacuum chamber can help cut down on the scattered radiation. One problem with this approach is that windows form a significant part of the interior surface area (30% and 50% for RC and SPA chambers, respectively). Thus a lot of the scattered power would find its way out of the chambers, where it can get reflected from the metallic lab equipment, and return back into the chamber through the same windows.

To define the problem more precisely, consider Fig. 106 showing a quarter of the SPA vacuum chamber. (To speed up computation, we can apply suitable boundary conditions based on the symmetry of the problem, and need only solve for the fields in one quarter of the volume.) The microwave beam will enter through the window on the top and leave through the window on the bottom of the chamber, passing between two ring-shaped electrodes (only

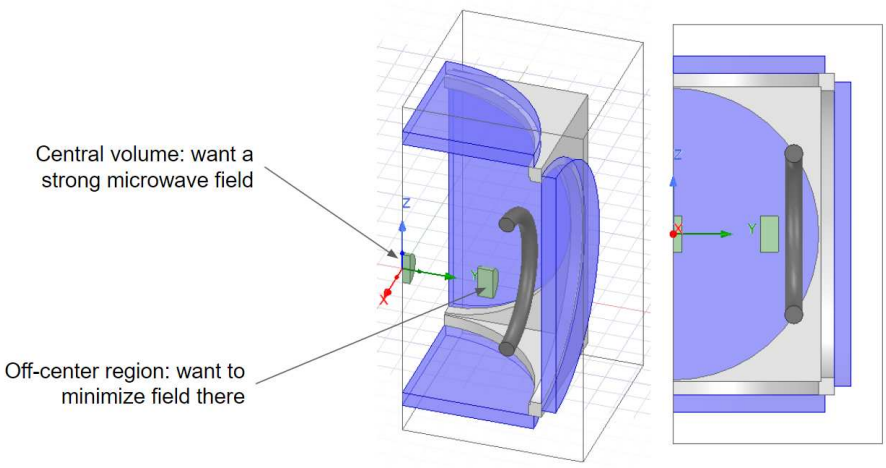

Figure 106: A quarter-section of the SPA vacuum chamber with ring electrodes. The two green regions highlight the areas where we do and do not want a significant amount of microwave radiation.

half of one ring shown in the figure for symmetry reasons). We desire a strong microwave field in the central region, highlighted by the left green volume, and no microwaves in the off-center green region to the right.

The magnitude of the microwave electric field is shown qualitatively in Fig. 107 on a logarithmic scale. The simulation is repeated several times. In the leftmost figure, only the microwave beam is included in the simulation. In the figures to the right, various experimental features are included in order to study their effect on scattering the microwave beam. Qualitatively, it is obvious that the windows have the largest effect in disturbing the beam.

In particular, observe that the presence of the vacuum chamber only contributes minimally to the scattering. Intuitively, covering the interior walls of the vacuum chamber with a perfect absorber would be equivalent to making the vacuum chamber effectively disappear from the problem. However, since the chamber is such a small contributor to the scattering, the effect of applying the absorber appears to be negligible, even for an absolutely perfect absorber. This is borne out by the quantitative HFSS simulation results in Table VIII, where it is seen that the presence or absence of the vacuum chamber does not significantly affect the scattered microwaves.

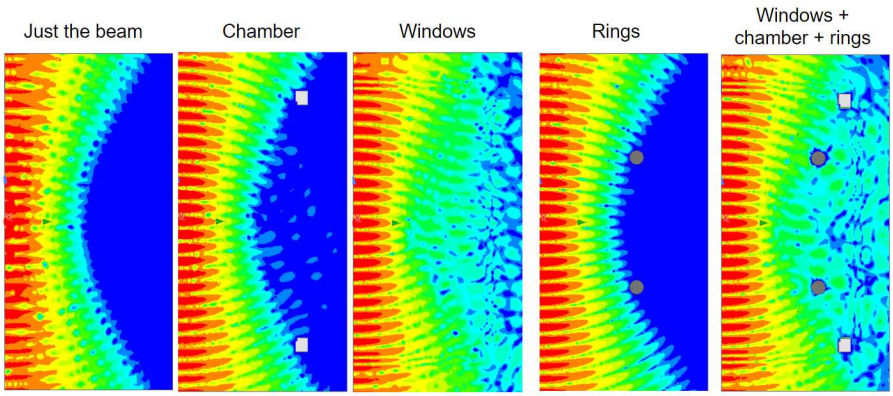

Figure 107: Magnitude of microwave electric field on a log scale. For qualitative explanation only. The figures show half of a cut through the center of the SPA vacuum chamber.

just the beam	0.10%
just chamber	0.09%
just windows	0.98%
just rings	0.11%
chamber + rings	0.11%
win + chamber	0.97%
win + rings	1.03%
win + chamber + rings	1.02%

Table VIII: The ratio, expressed in percent, of the average magnitude of the electric field in the off-center region compared to the center region. The two regions are defined in Fig. 106.